NyayaLens: A Local, Evidence-Linked Pipeline for Reading Indian Legal Agreements

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***Abstract*—NyayaLens reads an Indian legal agreement (a rental, licence, land lease, sale, builder-buyer, employment or non-disclosure agreement) and returns the parties, amounts, dates, obligations and clauses it contains, a set of review flags, and answers to questions about the text. Each output carries the character span it came from. The system runs on one machine and calls no external AI service. Extraction, clause labelling, risk rules and question answering are rule-based; a transformer baseline and a fine-tuned legal language model were tried and did not become part of the pipeline. We describe the system, a synthetic dataset of 3,640 annotated documents used to evaluate it, and the results. On held-out synthetic data, entity F1 is 0.994, clause F1 0.985 and relation F1 0.986; actor-and-action obligation F1 is 0.914. These numbers measure agreement with a generator written by the same team as the rules, so they show that the rules handle the wording they were built for, not how they would do on real agreements. We also report what happened when four hand-written-style agreements were run through the system: it first missed every party name and date and raised a false liability warning, and we describe the fixes. We list the limits plainly, including a failed fine-tuning experiment.**

***Index Terms*—legal document analysis, named entity recognition, rule-based extraction, synthetic data, retrieval, knowledge graph, Indian law**

I. INTRODUCTION

Residential and commercial agreements in India are long, repetitive and often drafted from templates that people edit by hand. The terms that matter to a signer (a deposit, a notice period, who bears repairs) sit in ordinary clauses, and a contradiction between two clauses is easy to miss. Reading aids exist for English-language contracts [10], and legal-domain language models exist for Indian court text [4][5], but we did not find an open tool aimed at Indian private agreements that works offline and shows its evidence. NyayaLens is a prototype built to explore that gap. It is a student project, and we do not claim it is the first or the best tool of its kind.

II. PROBLEM STATEMENT AND MOTIVATION

Given the text of an agreement and its declared type, the system should (a) extract typed values and the clauses they sit in, (b) state who must do what, (c) flag provisions and inconsistencies that deserve a second look, and (d) answer factual questions, refusing when the document does not say. Two constraints shaped the design. First, the text of a private agreement is sensitive, so nothing should leave the machine. Second, a reader has to be able to check every claim, so each output must point to its source text. These constraints favour transparent rules over a model whose errors cannot be traced to a span.

III. RELATED WORK

Transformer encoders [1] pre-trained as BERT [2] are the standard base for token classification, and the Hugging Face library [12] is the usual way to run them. Legal-domain versions include LEGAL-BERT [3] and InLegalBERT [4], a model for Indian law. Kalamkar et al. built a named-entity dataset of Indian court judgments [5]. Court judgments differ from private agreements in vocabulary and structure, so these resources do not transfer directly. For contracts, CUAD provides expert annotations of clause types in commercial English contracts [10], and LexGLUE collects several English legal benchmarks [16]. Named-entity evaluation follows the CoNLL-2003 convention of exact match on span and type [9]; SemEval-2013 Task 9 defined strict and partial-overlap variants [11], which we use for one comparison. Retrieval uses BM25 [6]. Retrieval-augmented generation [7] pairs retrieval with a generator; our default answerer is extractive and our optional local generator only rephrases retrieved text. Knowledge graphs are surveyed in [8]. Near-duplicate checking in the dataset uses MinHash resemblance [13], and OCR uses Tesseract [14].

IV. OBJECTIVES AND CONTRIBUTIONS

- A document-scoped design in which each agreement type is described by a data file (a "DomainPack") rather than by code, with seven domains and thirteen document types registered.
- An extraction and analysis pipeline whose every output carries a text span.
- A generator for annotated synthetic agreements with held-out templates, adversarial and control pairs, counterfactual pairs and a question benchmark, with a validator for leakage and duplicates.
- An evaluation report produced by a script from the same code that runs in the application, including negative results.

V. SYSTEM ARCHITECTURE

A Next.js web application talks to an Express API backed by PostgreSQL. An uploaded file passes through stages that are recorded one by one, so a failure in an optional stage does not stop the others: text extraction (PDF or DOCX), OCR for pages without text, domain detection, rule-based entity extraction, clause segmentation and labelling, relation and obligation extraction, an optional transformer baseline, and the risk and consistency checks. Results are stored with character offsets into one canonical text. A knowledge graph is built from the stored rows when requested. Access to every document endpoint is checked in the same database query that loads the document. Passwords are hashed with Argon2id, access tokens are short-lived with rotating refresh tokens, cross-origin access is limited to configured origins, and failed sign-ins are rate-limited. Fig. 1 shows the stages.

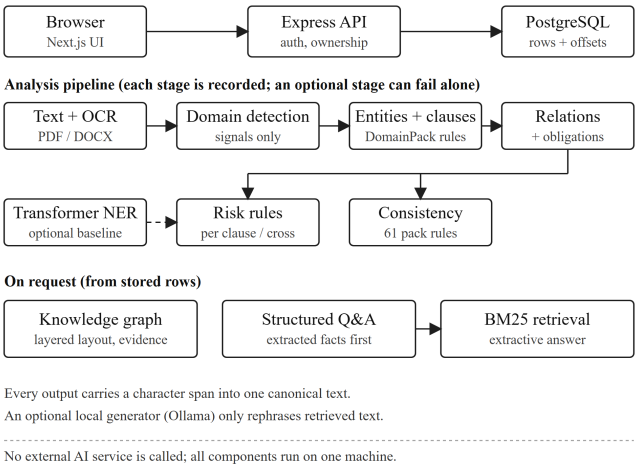


Fig. 1. System architecture. The dashed arrow marks an optional stage.

VI. DATASET CONSTRUCTION AND ANNOTATION

The generator (`research/generator/v3`) builds each document from clause templates and value pools and records the exact span of every annotation as it writes the text, so labels are not added afterwards by hand. The final dataset (seed 2027) has 3,640 documents, 520 per domain: 1,470 train, 315 dev and 315 test, plus 560 adversarial documents, 560 matched controls, 280 documents in 140 counterfactual pairs, and 140 cross-domain documents with generic titles. About 45,000 questions are generated with answers and evidence spans (45,275 in total). Test documents use held-out templates on half of their clauses. A validator checks for exact and near duplicates (MinHash with a Jaccard threshold of 0.8 [13]), invalid spans, and template or text leakage between train and test; the run reports zero of each (`research/data/reports/dataset_quality.json`). Adversarial documents contain one injected problem and each has a control twin made from the same seed without it. In a counterfactual pair, one value differs. Table I gives the split sizes.

TABLE I
COMPOSITION OF THE SEED-2027 DATASET

Split	Documents
Train	1,470
Development	315
Test	315
Adversarial	560
Matched controls	560
Counterfactual (140 pairs)	280
Cross-domain	140
Total	**3,640**

VII. METHODS FOR EXTRACTION

A. Entities

A DomainPack lists, for each domain, the roles that exist (for example BUYER, ADVANCE_AMOUNT, NOTICE_PERIOD), each mapped to one of twelve core types. Values (money, dates, durations, percentages, areas) are found by patterns and given a role by the words around them. Party names are found from definitions in the text. Across the seven packs there are 144 roles. Typographic quotes are swapped for plain ones one character for one character, so offsets stay valid.

B. Clauses

A segmenter recognises six heading styles. Each clause is labelled from its heading using the pack's ordered patterns. There are 124 clause types across the packs.

C. Normalisation

Amounts become rupee integers (including lakh and crore forms), dates become ISO dates, durations become days, and areas keep their unit, so "Rs. 25,000" and "INR 25000" compare equal.

D. Relations and obligations

Relations follow the pack's schema (110 types). An obligation is a sentence with a modal verb ("shall", "must"), a party term and a verb from the pack's list; the record holds the actor, action, amount, deadline, frequency and condition. A sentence never crosses a clause boundary or swallows a heading. An actor the pack does not expect for that action is marked for review rather than corrected.

E. Domain detection

The declared type is compared with signals in the title and body. The result is a count of signals with no probability attached; a mismatch is shown to the user and never changes the stored type.

VIII. RISK DETECTION AND CONSISTENCY ANALYSIS

Per-clause rules look for discretion left to one party, structural repairs placed on the occupant, uncapped liability, post-employment restrictions and one-sided remedies. Cross-clause rules compare notice periods and look for expected clauses that are absent. A separate consistency engine applies each pack's rules (61 in all): one field should have one value, instalments should add to a total, dates should be in order, and an obligation's actor should fit the pack. Findings are worded as a potential issue for review, and a test rejects phrases such as "is illegal" or "is void". Findings are not legal conclusions.

IX. KNOWLEDGE GRAPH

The graph is computed on request from stored rows. Nodes are the document, clauses, entities, obligations, deadlines, risk findings and legal references; edges are structural containment, typed relations, "must perform", "due on", "contradicts", and "relevant to" for legal references. The legal-reference edge never says a law applies. Reference metadata is marked unverified. The interface opens on the entities that take part in a relation, duty or finding, because drawing every containment edge produced an unreadable tangle; clauses and the document node can be switched on. A layered layout orders nodes to reduce edge crossings, and selecting a node shows its source text and connections.

X. RETRIEVAL AND QUESTION ANSWERING

A question is first matched against the pack's field names and question words ("who" for a party, "how much" for money) and answered from extracted facts, with the spans as evidence. This covers single fields, two fields, differences between two amounts, and "is X stated consistently" questions. A question about a field that belongs to a different kind of document ("probation period" for a sale) is refused. Other questions go to BM25 retrieval [6] over clauses with an extractive answer. Every answer is marked grounded or not sufficient and lists the clause and page of each source. Retrieved text is treated as data, not instructions, and a set of prompt-injection cases is part of the tests.

XI. EXPERIMENTAL SETUP

Evaluation is run by `apps/api/scripts/evaluate-v3.mjs`, which calls the same functions as the application on each document's raw text, in memory; OCR is not exercised. Rules were developed against two other datasets (seeds 2026 and 99). The seed-2027 dataset was generated afterwards and was not inspected before its first evaluation. The seed-2026 test split was inspected during development, which we disclose because it weakens that split as evidence. Earlier phases used a 300-document set (150 rental, 150 land lease) with five-fold cross-validation grouped by document.

XII. METRICS

Entities are matched on exact span and label; clauses on type and overlapping span; relations on subject role, relation type and the normalised object value. Precision is TP/(TP+FP), recall TP/(TP+FN), F1 their harmonic mean; scores are micro-averaged. For the transformer baseline we also report span-overlap F1 with a match at intersection-over-union of at least 0.5, in the manner of partial matching in [11]. Question answering is scored as correct when the answer contains the expected value and, for unanswerable questions, when the system refuses; evidence is correct when a cited span overlaps a true mention of the asked field.

XIII. RESULTS

A. Extraction

Seed 2027; `docs/evaluation.md`. Table II lists the scores.

TABLE II
EXTRACTION RESULTS ON THE SEED-2027 DATA (MICRO-AVERAGED F1)

Split (documents)	Entity F1	Clause F1	Relation F1	Obligation F1
Test (315)	0.994	0.985	0.986	0.914
Cross-domain (140)	0.994	0.987	0.986	0.899

Obligation recall is lower than precision (about 0.87 against 0.96 on the test split): duties stated over several sentences or without a modal verb are missed. Domain detection names the right domain for all test and cross-domain documents; the exact document type is right for 96.5% of test documents and 76.4% of cross-domain documents, whose titles are generic. Fig. 2 plots the four scores for both splits.

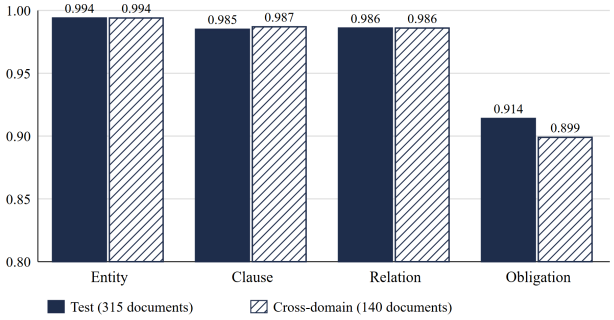


Fig. 2. F1 on the seed-2027 test and cross-domain splits. The vertical axis starts at 0.80.

B. Consistency

Injected inconsistencies were detected in 100% of cases for seven of nine categories and 110/120 for duration contradictions, with the cited span correct in each detected

case. Control documents raised no false finding except four of 33 for property identifiers. Of 315 ordinary test documents, 10 (3.2%) had a consistency finding.

C. Risk rules

Ordinary test documents. Discretion (11 cases), asymmetric notice (17), one-sided remedy (9), restrictive covenant (5), uncapped liability (3) and land-use contradiction (6) were found with precision and recall of 1.0 on these counts. Structural-repair liability shift was found in 2 of 3 cases. Missing-clause findings had precision 0.885 and recall 1.0 (69 true, 9 false positives, all for the intellectual-property clause of employment contracts). Counts are small, so these are indicative only. Table III summarises these counts.

TABLE III
RISK-RULE RESULTS ON ORDINARY TEST DOCUMENTS (COUNTS ARE SMALL)

Category	Cases	Precision	Recall
Discretion	11	1.00	1.00
Asymmetric notice	17	1.00	1.00
One-sided remedy	9	1.00	1.00
Restrictive covenant	5	1.00	1.00
Uncapped liability	3	1.00	1.00
Land-use contradiction	6	1.00	1.00
Repair liability shift	3	n/a	0.67
Missing clause	69 + 9 FP	0.885	1.00

D. Counterfactuals

261 of 262 one-value changes were reflected in the extracted values.

E. Question answering

Answer accuracy, evidence equal. Factual 99.4% (15,549 questions), unanswerable 99.5% (2,459), cross-clause 100% (883), contradiction 98.8% (1,120), multi-hop 100% (357). The questions are generated, with the generator's phrasing. Fig. 3 shows these values.

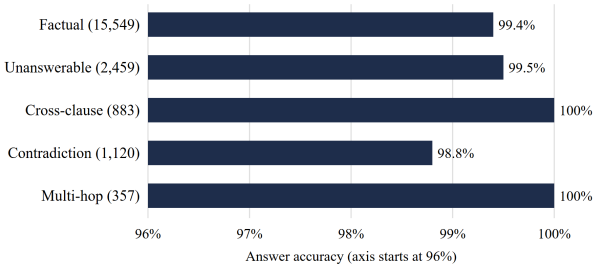


Fig. 3. Answer accuracy by question type on the generated benchmark (item counts in parentheses). The axis starts at 96%.

F. Earlier retrieval evaluation

Phase 7A, 33 hand-written cases, `apps/api/rag_eval`. The right clause was among the top three in 28 of 28 answerable cases for at least one expected clause and in 27 of 28 for all of them; 96.4% of answerable questions were grounded; 88.9% of grounded answers (24 of 27) cited the right evidence; 4 of 5 unanswerable questions were refused. The cases were written by the same team as the system.

G. Earlier phases

On the 300-document set, the rule-based extractor scored 1.0 F1 on seven labels, 0.507 on land area (cause not confirmed) and 0 on land-use restriction, for which no pattern was written. The transformer baseline (a distilled BERT NER model [15]) scored 0 under label-exact matching because its label set differs, and F1 0.247 (precision 0.210, recall 0.299) under label-agnostic span overlap, so its spans were also weak for this text. A fine-tuning run of InLegalBERT [4] was limited by CPU-only hardware to one epoch on 60 training documents per fold; it produced no entities at all. This is a negative result of the compute available, not evidence about the model. On a 75-document adversarial set from Phase 6, all five risk categories were detected 15 of 15 times with no control flagged; that set is separate from the v3 data above.

H. Four hand-written-style agreements

We ran four synthetic agreements written in a different style from the generator (a rental, a leave and licence, a land lease and a sale agreement) through the application. The first run found no party name and no date in any of them, dropped the 60-day and 30-day notice periods, truncated a survey identifier (`884A` read as `884`), raised a false liability-shift flag on two documents (a clause where the owner arranges repairs and excepts damage the occupant causes was read as putting repairs on the occupant), and missed the planted unfair provision in three of the three documents that had one. The causes were rules written only for the generator's wording: party definitions of the form `(the "Landlord")`, dates like `1 November 2026`, short headings such as `TERM`, and discretion and termination phrasings. We fixed these with general rules, and checked on a separate seed-99 set that no benchmark score fell (57 of 64 metrics unchanged, seven improved, none worse). After the fixes all expected parties, dates, amounts and identifiers were found, the false flags were gone, and the three planted provisions were flagged with the correct clause as evidence. Because these four documents guided the fixes, they now demonstrate the fixes and are not independent evidence (`apps/api/tests/sampleAgreements.test.ts`).

XIV. ERROR ANALYSIS

The largest gap is between generator wording and ordinary wording, shown by the four-document run: a score of 0.99 on generated text coexisted with missing every party name on hand-written text. Within the benchmark, remaining errors are mostly in obligations spread over several sentences, a clause with an unusual heading (which

is then missed and also produces a false "missing clause" finding), a structural-repair clause worded in a way the rule does not cover, and four property-identifier controls that were flagged although nothing was injected. The retrieval fallback has no way to tell a topically close passage from an answer.

XV. DISCUSSION

Transparent rules gave traceable output and no training cost, and a rule can be fixed when it fails, as the four-document run showed. The price is coverage: each new wording needs a rule. The held-out-template split reduces but cannot remove the circularity of testing rules on a generator written by the same people. The more informative result of this project may be the hand-written run, because it exposed defects that the benchmark could not.

XVI. LIMITATIONS AND THREATS TO VALIDITY

- All evaluation is synthetic and written by the team that wrote the rules. Real-agreement accuracy is unmeasured.
- The seed-2026 test split was inspected during development; the seed-2027 split was used for the reported numbers and was not tuned on, but the generator's templates and value pools are shared across seeds.
- Question-answer items use the generator's phrasing, so free-form questions are not covered.
- Per-category risk counts are small.
- No model was trained for the reported results. The optional transformer baseline performed poorly and the fine-tuning run produced nothing usable.
- English only; OCR quality is not evaluated.
- Legal-reference metadata is unverified and is never used to produce a finding.
- Not every domain has been validated on real documents.

XVII. ETHICAL, PRIVACY AND LEGAL CONSIDERATIONS

Documents are processed on the user's machine and stored in a local database. The output is a reading aid, not legal advice, and the interface says so. Flags are worded as items for review. Synthetic names and addresses are invented. We did not use real agreements. Before any public deployment, a lawyer should review the wording and the unverified legal references, the cross-origin and rate-limit settings should be reviewed, and real documents should be handled under a retention policy.

XVIII. FUTURE WORK

A small consented set of real, anonymised agreements for honest evaluation; training an entity model on the generated data and comparing it with the rules; local embeddings for retrieval; Hindi and mixed-language documents; and lawyer-verified statutory references.

XIX. CONCLUSION

NyayaLens shows that a rule-based, offline, evidence-linked reader for Indian agreements is feasible and can score highly on generated data. Its main weakness is generalisation, which we measured on four hand-written agreements and partly repaired. Whether the approach holds on real agreements is the open question for the next stage.

REFERENCES

[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser and I. Polosukhin, "Attention Is All You Need," arXiv:1706.03762, 2017. https://arxiv.org/abs/1706.03762

[2] J. Devlin, M.-W. Chang, K. Lee and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," arXiv:1810.04805, 2018. https://arxiv.org/abs/1810.04805

[3] I. Chalkidis, M. Fergadiotis, P. Malakasiotis, N. Aletras and I. Androutsopoulos, "LEGAL-BERT: The Muppets straight out of Law School," Findings of EMNLP 2020; arXiv:2010.02559. https://arxiv.org/abs/2010.02559

[4] S. Paul, A. Mandal, P. Goyal and S. Ghosh, "Pre-trained Language Models for the Legal Domain: A Case Study on Indian Law," 19th International Conference on Artificial Intelligence and Law (ICAIL 2023); arXiv:2209.06049. https://arxiv.org/abs/2209.06049

[5] P. Kalamkar, A. Agarwal, A. Tiwari, S. Gupta, S. Karn and V. Raghavan, "Named Entity Recognition in Indian court judgments," NLLP Workshop 2022; arXiv:2211.03442. https://arxiv.org/abs/2211.03442

[6] S. Robertson and H. Zaragoza, "The Probabilistic Relevance Framework: BM25 and Beyond," Foundations and Trends in Information Retrieval, vol. 3, no. 4, pp. 333-389, 2009. doi:10.1561/1500000019

[7] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel and D. Kiela, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," NeurIPS 2020; arXiv:2005.11401. https://arxiv.org/abs/2005.11401

[8] A. Hogan et al., "Knowledge Graphs," ACM Computing Surveys, vol. 54, no. 4, article 71, 2021. doi:10.1145/3447772; arXiv:2003.02320.

[9] E. F. Tjong Kim Sang and F. De Meulder, "Introduction to the CoNLL-2003 Shared Task: Language-Independent Named Entity Recognition," Proc. CoNLL-2003, Edmonton, pp. 142-147; arXiv:cs/0306050. https://arxiv.org/abs/cs/0306050

[10] D. Hendrycks, C. Burns, A. Chen and S. Ball, "CUAD: An Expert-Annotated NLP Dataset for Legal Contract Review," NeurIPS 2021 (Datasets and Benchmarks); arXiv:2103.06268. https://arxiv.org/abs/2103.06268

[11] I. Segura-Bedmar, P. Martínez and M. Herrero-Zazo, "SemEval-2013 Task 9: Extraction of Drug-Drug Interactions from Biomedical Texts (DDIExtraction 2013)," Proc. SemEval-2013, Atlanta, pp. 341-350. https://aclanthology.org/S13-2056

[12] T. Wolf, L. Debut, V. Sanh et al., "HuggingFace's Transformers: State-of-the-art Natural Language Processing," arXiv:1910.03771, 2019. https://arxiv.org/abs/1910.03771

[13] A. Z. Broder, "On the resemblance and containment of documents," Proc. Compression and Complexity of Sequences, 1997. https://www.cs.princeton.edu/courses/archive/spring13/cos598C/broder97resemblance.pdf

[14] R. Smith, "An Overview of the Tesseract OCR Engine," Proc. Ninth International Conference on Document Analysis and Recognition (ICDAR), 2007, pp. 629-633. https://ieeexplore.ieee.org/document/4376991

[15] V. Sanh, L. Debut, J. Chaumond and T. Wolf, "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter,"

arXiv:1910.01108, 2019. https://arxiv.org/abs/1910.01108

[16] I. Chalkidis, A. Jana, D. Hartung, M. Bommarito, I. Androutsopoulos, D. M. Katz and N. Aletras, "LexGLUE: A Benchmark Dataset for Legal Language Understanding in English," ACL 2022; arXiv:2110.00976. https://arxiv.org/abs/2110.00976

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